

Evidence-First TBML Corridor Drift Lab - Final analytical report

This output prioritises an unusual corridor-product pattern for human review. It does not establish money laundering, misinvoicing, or criminal intent.

Executive summary

Purpose: Summarize what the official-data lab built and what it found.

This package implements a simplified official-data analytical lab using a filtered official-derived CEPII BACI HS17 V202601 Parquet extract, the World Bank Commodity Markets annual workbook, time-safe features, scenario-based model evaluation, evidence rows, FATF-Egmont typology cards, and grounded analyst briefs.

The official BACI extract contains **25,844 retained panel rows** after row-quality assignment. The panel covers **2017-2024** and exactly three HS6 families: crude palm oil 151110, refined copper cathodes 740311, and non-monetary unwrought gold 710812.

The retained review-priority scoring approach is `hybrid_score` using challenger `xgboost` with validation-selected challenger weight `0.75`. Scores are ranking signals, not probabilities of crime.

Source feasibility and verification

Purpose: Show that the project ran on the uploaded official or official-derived files and did not substitute synthetic trade data.

Source file inventory

file_name	size_bytes	sha256
baci_hs17_v202601_selected_2017_2024_151110_740311_710812.parquet	340203	0e493e057c8fd57902328551a9c874d22aea0e4c3437432617489c7c3571fe12
data_source_notes.json	4342	a8a973d0797cbfcd4064b1dfd071faa441bdc7b766e719af9bbb53014365bfe6
CMO-Historical-Data-Annual.xlsx	3177955	

file_name	size_bytes	sha256
		9fbc348f40ecdb02eb1bcf858a2965383d2aaf3445e8920dd7d60ae7b04af51
Trade-Based-Money-Laundering-Trends-and-Developments_2020.pdf	3108080	0969e49877a1937a7bc49ac02ab10a444adca2c2ff2c7803ee6a35adea9f6828
Trade-Based-Money-Laundering-Risk-Indicators_2021.pdf	2079148	670b80270b33354b17263d44393cd65386a96bdb1a171365f217048e7cca6afb
country_codes_V202601.csv	5346	c6892f28bea23d689b29b074d299cea7f5196c708b03026fbe419fa4ecb4e820
product_codes_HS17_V202601.csv	598831	4abb74d330c3a1969ccfc4652e6ef849963980d55e45ad768eb54a32747efd5e
baci-readme.txt	571	156c8fe9d8582f24fdec5c43d912e3736dc515a6efd5cd04e513095a0db44c66

Verification summary

item	status	detail
BACI row count	pass	25844
BACI years	pass	{'2017': 2783, '2018': 3094, '2019': 3279, '2020': 3162, '2021': 3358, '2022': 3510, '2023': 3445, '2024': 3213}
BACI HS6	pass	{'151110': 6717, '710812': 12205, '740311': 6922}
World Bank annual benchmarks	pass	{'crude_palm_oil': 'USD per metric ton', 'gold_unwrought': 'USD per troy ounce', 'refined_copper_cathodes': 'USD per metric ton'}
FATF-Egmont PDFs	pass	[{'file_name': 'Trade-Based-Money-Laundering-Trends-and-Developments_2020.pdf', 'pages': 66, 'readable': True, 'sample_text_length_first_pages': 3461}, {'file_name': 'Trade-Based-Money-Laundering-Risk-Indicators_2021.pdf', 'pages': 10, 'readable': True, 'sample_text_length_first_pages': 4408}]
Web check	pass	Web search found official FATF pages for the 2020 trends report and 2021 risk indicators; no newer official TBML-specific replacement was identified during this run.

BACI  is interpreted as thousands of current USD and multiplied by 1,000. BACI  is interpreted as metric tons. HS6  is treated as a six-character string.

HS6 family coverage

hs6	family_id	product_name	panel_rows
151110	crude_palm_oil	Crude palm oil	6717

hs6	family_id	product_name	panel_rows
710812	gold_unwrought	Non-monetary unwrought gold	12205
740311	refined_copper_cathodes	Refined copper cathodes	6922

Row quality status

quality_status	rows
fully_usable	11894
usable_with_caveat	11762
excluded_from_modeling_retained_for_audit	2188

Official-data pipeline architecture

Purpose: Show the actual simplified architecture implemented in this package.

Official Data Pipeline Architecture

Graphviz 'dot' was not available; see .dot source for diagram.

Official-data pipeline architecture

Source To Report Data Flow

Graphviz 'dot' was not available; see .dot source for diagram.

Data flow from source files to final outputs

World Bank benchmark handling

Purpose: Explain benchmark extraction and unit normalization.

Palm oil and copper benchmarks are preserved as USD per metric ton. Gold is preserved in its original USD per troy ounce unit and converted to USD per metric ton using `gold_usd_per_metric_ton = gold_usd_per_troy_ounce * (1_000_000 / 31.1034768)`.

family_id	hs6	benchmark_year	benchmark_price_original	benchmark_unit_original	benchmark_price_usd_per_metric_ton	benchmark_conversion_method
crude_palm_oil	151110	2017	750.81	USD per metric ton	750.81	identity
crude_palm_oil	151110	2018	638.66	USD per metric ton	638.66	identity
crude_palm_oil	151110	2019	601.37	USD per metric ton	601.37	identity
crude_palm_oil	151110	2020	751.77	USD per metric ton	751.77	identity
crude_palm_oil	151110	2021	1130.58	USD per metric ton	1130.58	identity

family_id	hs6	benchmark_year	benchmark_price_original	benchmark_unit_original	benchmark_price_usd_per_metric_ton	benchmark_conversion_method
crude_palm_oil	151110	2022	1275.99	USD per metric ton	1275.99	identity
crude_palm_oil	151110	2023	886.45	USD per metric ton	886.45	identity
crude_palm_oil	151110	2024	963.36	USD per metric ton	963.36	identity
gold_unwrought	710812	2017	1257.56	USD per troy ounce	4.04315e+07	gold_usd_per_troy_ounce * (1000000 / 31.1034768)
gold_unwrought	710812	2018	1269.23	USD per troy ounce	4.08067e+07	gold_usd_per_troy_ounce * (1000000 / 31.1034768)
gold_unwrought	710812	2019	1392.5	USD per troy ounce	4.47699e+07	gold_usd_per_troy_ounce * (1000000 / 31.1034768)
gold_unwrought	710812	2020	1770.25	USD per troy ounce	5.69149e+07	gold_usd_per_troy_ounce * (1000000 / 31.1034768)

Time-safe features

Purpose: Build review signals without letting current rows define their own history or using future years.

Graphviz 'dot' was not available; see .dot source for diagram.

Time-safe feature construction flow

Time-safe feature explanation table

feature_name	derivation	why_created	significance_for_review	time_safety_rule	plain_english_interpretation
log_unit_value	log(unit_value_usd_per_metric_ton)	Stabilizes skewed unit values.	Shows unusually high or low implied prices within a product family.	Uses only the current row's official value and quantity.	The aggregate implied price on a log scale.
shifted_corridor_history_median	Median of prior-year log_unit_value for the same exporter-importer-HS6 corridor.	Builds the corridor's own baseline.	Shows whether the row differs from its own past.	Prior years only; current/future years excluded.	What this route-product normally looked like before this year.
shifted_corridor_history_mad	Median absolute deviation of prior prior-year log_unit_value.	Gives robust historical spread.	Large current deviations matter more when history is stable.	Prior years only.	How much this route-product used to vary.
robust_historical_z	(log_unit_value - shifted median) / shifted MAD with a safe denominator.	Measures distance from prior corridor behavior.	Core drift signal.	Median and MAD are shifted prior-year values.	How far this row is from its own historical pattern.
same_family_year_peer_percentile	Percentile rank of benchmark_residual within the same	Compares against contemporaneous peers.	Finds rows unusual relative to other corridors in the same market year.	Same-year only; no future-year peer distributions.	Where this row sits among same-product peers that year.

feature_name	derivation	why_created	significance_for_review	time_safety_rule	plain_english_interpretation
	HS6/family and year.				
trade_value_yoy_change	log(current trade value / previous observed trade value) for same corridor-HS6.	Captures sudden value movement.	Highlights sharp trade value jumps or falls.	Previous observation only.	How much aggregate value changed since the previous active year.
quantity_yoy_change	log(current quantity / previous observed quantity) for same corridor-HS6.	Separates value growth from physical quantity growth.	Shows whether value changes are volume-driven.	Previous observation only.	How much physical quantity changed.
unit_value_yoy_change	log(current unit value / previous observed unit value).	Captures implied price movement.	Important for valuation-style review.	Previous observation only.	How much the implied price changed.
benchmark_residual	log_unit_value - log(World Bank benchmark USD per metric ton).	Adds macro commodity context.	Separates corridor-specific gaps from broad commodity levels.	Uses benchmark for the same year only.	The gap between observed aggregate unit value and broad market context.
benchmark_adjusted_drift	benchmark_residual - previous observed benchmark_residual.	Looks at movement after broad benchmark changes.	Reduces false positives when commodity markets move broadly.	Previous residual only.	Whether the gap to benchmark changed compared with this route's prior gap.
corridor_activity_history	Count of prior active years for exporter-importer-HS6.	Captures history depth.	Short-history rows need caveats.	Prior years only.	How many earlier years this route-product appeared.
corridor_novelty_flag	1 if no prior active year exists; else 0.	Flags new corridors.	New corridors can matter when paired with extreme values.	Prior activity only.	This route-product is new in the observed panel.
corridor_reactivation_flag	1 if prior activity exists and the previous active year is not the immediately preceding year.	Captures reappearing routes.	Reactivation can matter with large valuation changes.	Prior activity pattern only.	The route-product disappeared and came back.
value_quantity_divergence	trade_value_yoy_change - quantity_yoy_change.	Identifies value moving faster than quantity.	Useful for valuation-style review patterns.	Based on previous-observation changes only.	Value changed more than physical volume.
missingness_flags	Binary flags for missing quantity, benchmark, history, or country mapping.	Makes data limits explicit.	Prevents data-quality issues from being mistaken for suspiciousness.	Missingness reflects available data at the row time.	What key information is missing.
data_quality_flags	Flags and additive score for quantity, benchmark, history, and	Separates confidence from anomaly strength.	Low quality caveats a case; it should not automatically	Uses source/provenance and prior history only where relevant.	How usable and well-supported the row is.

feature_name	derivation	why_created	significance_for_review	time_safety_rule	plain_english_interpretation
	provenance completeness.		raise suspiciousness.		

Scenario layer and model comparison

Purpose: Evaluate ranking methods without pretending that public BACI contains confirmed TBML labels.

Train Validation Test ML Workflow

Graphviz 'dot' was not available; see .dotsource for diagram.

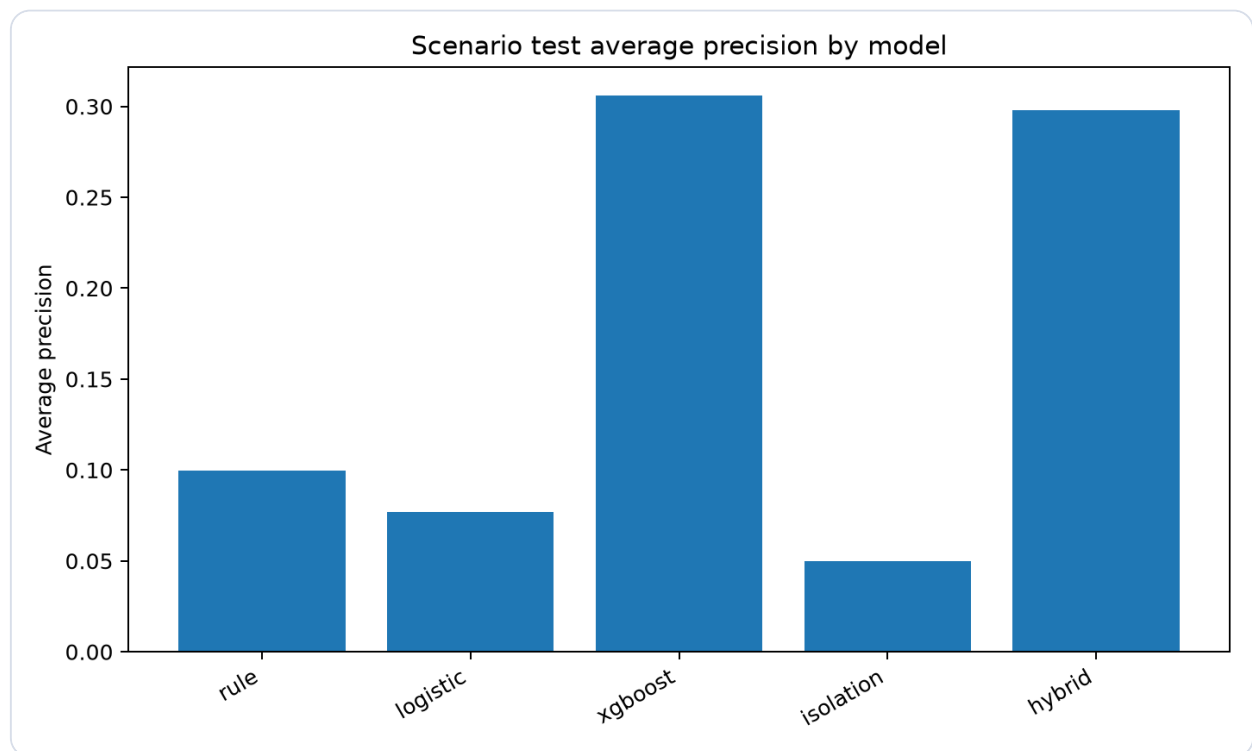
Train / validation / test ML workflow

Frozen split summary

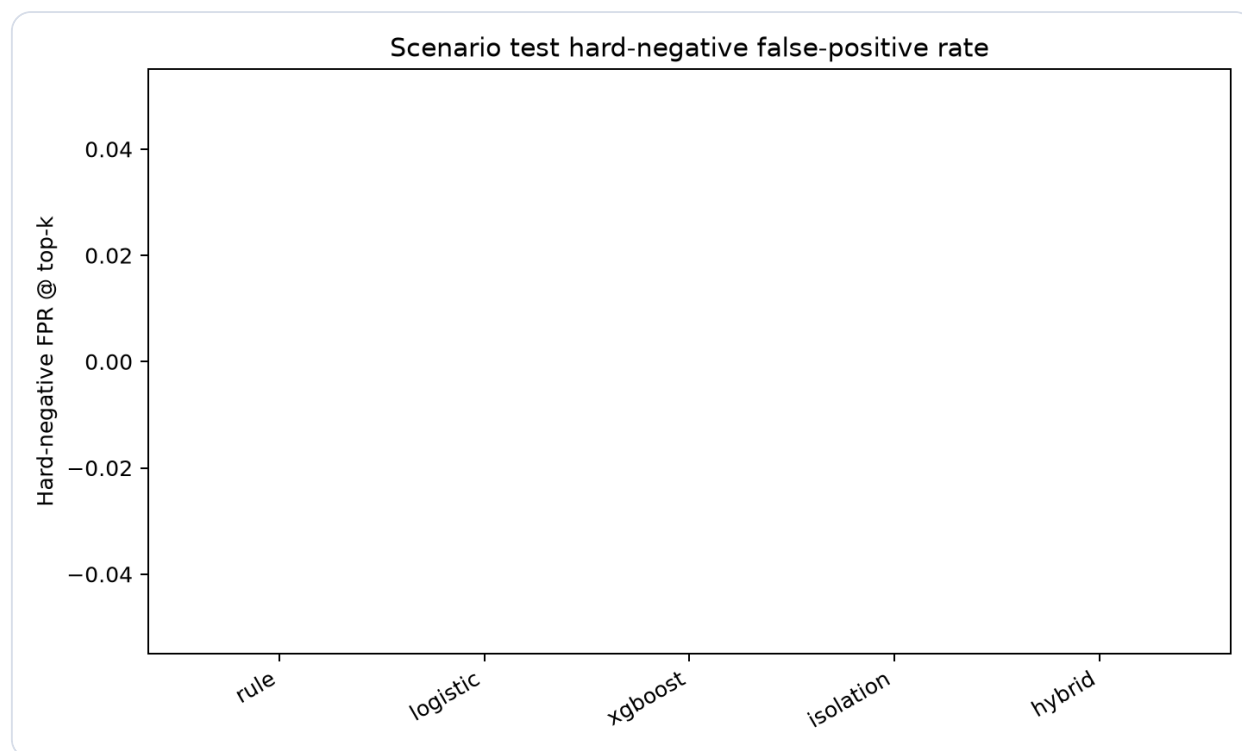
split	years	rows	synthetic_positive_rows	hard_negative_rows
train	2017, 2018, 2019, 2020	12318	108	72
validation	2021, 2022	6868	108	72
test	2023, 2024	6658	108	72

Model comparison - test split, all families

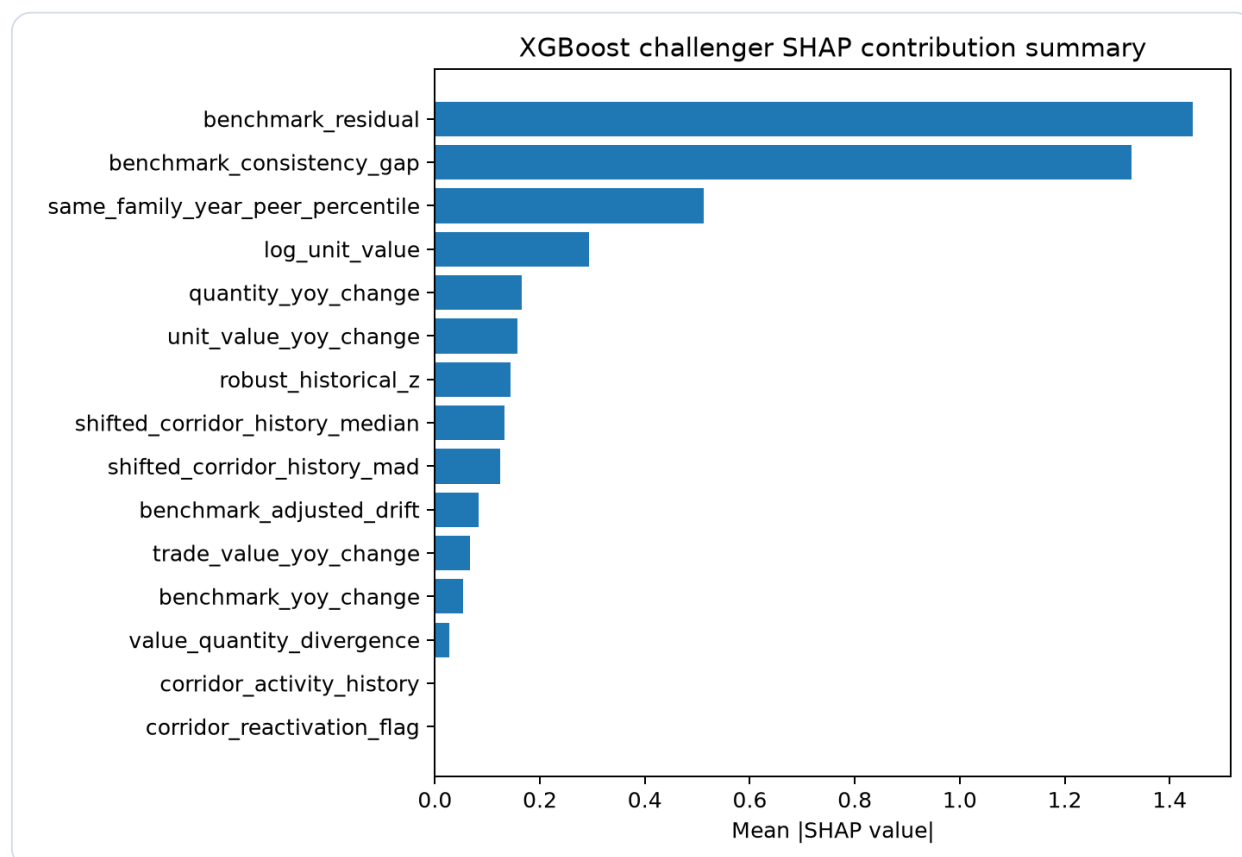
model	precision_at_k	recall_at_k	lift_at_k	average_precision	hard_negative_false_positive_rate	ordinary_false_positive_rate
xgboost	0.5	0.231481	27.9306	0.306143	0	0.00427131
hybrid	0.48	0.222222	26.8133	0.297849	0	0.00444217
rule	0.14	0.0648148	7.82056	0.0996167	0	0.00734666
logistic	0.06	0.0277778	3.35167	0.0770069	0	0.00803007
isolation	0.06	0.0277778	3.35167	0.0499403	0	0.00803007



Model comparison figure



Hard-negative false-positive comparison



SHAP summary - model contribution only

Top-ranked official observations

Purpose: Apply the selected scoring approach to real official observations only. Scenario rows are not used in this review queue.

rank	year	exporter_iso3	importer_iso3	hs6	product_name	selected_review_priority_score	quality_status	key_evidence_count
1	2022	ESP	NLD	710812	Non-monetary unwrought gold	0.989129	usable_with_caveat	4
2	2022	KWT	NPL	710812	Non-monetary unwrought gold	0.985756	usable_with_caveat	4
3	2022	QAT	NPL	710812	Non-monetary unwrought gold	0.98547	usable_with_caveat	4
4	2022	HKG	NPL	710812	Non-monetary unwrought gold	0.98547	usable_with_caveat	4
5	2022	SAU	NPL	710812	Non-monetary unwrought gold	0.984675	usable_with_caveat	4
6	2022	ITA	MKD	151110	Crude palm oil	0.980196	usable_with_caveat	4
7	2024	GBR	ITA	740311	Refined copper cathodes	0.980173	usable_with_caveat	4
8	2022	CAF	CHE	710812	Non-monetary unwrought gold	0.978825	fully_usable	4
9	2018	USA	IRL	740311	Refined copper cathodes	0.978747	usable_with_caveat	4
10	2020	ITA	TUR	740311	Refined copper cathodes	0.975341	usable_with_caveat	4
11	2022	SGP	JPN	740311	Refined copper cathodes	0.974686	usable_with_caveat	4
12	2024	CHN	GHA	740311	Refined copper cathodes	0.973245	usable_with_caveat	4
13	2019	IND	ITA	710812	Non-monetary	0.97265	usable_with_caveat	4

rank	year	exporter_iso3	importer_iso3	hs6	product_name	selected_review_priority_score	quality_status	key_evidence_count
					unwrought gold			
14	2020	BEL	SVK	740311	Refined copper cathodes	0.972575	usable_with_caveat	4
15	2019	NGA	IRL	151110	Crude palm oil	0.971741	usable_with_caveat	4
16	2022	DEU	BRA	740311	Refined copper cathodes	0.971565	usable_with_caveat	4
17	2022	MYS	NPL	710812	Non-monetary unwrought gold	0.971243	usable_with_caveat	4
18	2022	KOR	NPL	710812	Non-monetary unwrought gold	0.971243	usable_with_caveat	4
19	2022	JPN	PHL	710812	Non-monetary unwrought gold	0.970299	usable_with_caveat	4
20	2023	NLD	SGP	740311	Refined copper cathodes	0.969581	usable_with_caveat	4

Evidence and grounded briefs

Purpose: Convert scores into recomputable facts and bounded analyst-facing narratives.

Evidence Genai Grounded Brief Flow

Graphviz 'dot' was not available; see .dot source for diagram.

Evidence and grounded brief flow

Evidence table sample

evidence_id	obs_id	evidence_type	metric_name	observed_value	threshold	severity	plain_english_summary
ev_64fe9330ad36973b98	obs_c1178b54e91326a9b0f5	history_deviation	robust_historical_z	18.8267	4	high	2022 ESP to NLD 710812 (Non-monetary unwrought gold) differs from its prior corridor history with robust historical z of 18.827. Severity: high.
ev_89c989e7cf036ef947	obs_c1178b54e91326a9b0f5	benchmark_adjusted_drift	benchmark_adjusted_drift	1.85827	0.7	high	2022 ESP to NLD 710812 (Non-monetary unwrought gold) moved by 1.858 in benchmark-adjusted residual versus the previous

evidence_id	obs_id	evidence_type	metric_name	observed_value	threshold	severity	plain_english_summary
							observed corridor year. Severity: high.
ev_03b1acd36c34032643	obs_c1178b54e91326a9b0f5	benchmark_gap	benchmark_residual	1.8505	0.7	high	2022 ESP to NLD 710812 (Non-monetary unwrought gold) has a log benchmark residual of 1.850 against the World Bank annual benchmark. Severity: high.
ev_d25e9b9a9f2e0660b3	obs_c1178b54e91326a9b0f5	annual_unit_value_change	unit_value_yearly_change	1.8588	0.8	high	2022 ESP to NLD 710812 (Non-monetary unwrought gold) has a year-over-year log unit-value change of 1.859. Severity: high.
ev_9112748c22f2de5977	obs_31f1dbac37c15eb7186d	history_deviation	robust_historical_z	12.2572	4	high	2022 KWT to NPL 710812 (Non-monetary unwrought gold) differs from its prior corridor history with robust historical z of 12.257. Severity: high.
ev_451e6e42094546bc46	obs_31f1dbac37c15eb7186d	benchmark_adjusted_drift	benchmark_adjusted_drift	1.99783	0.7	high	2022 KWT to NPL 710812 (Non-monetary unwrought gold) moved by 1.998 in benchmark-adjusted residual versus the previous observed corridor year.

evidence_id	obs_id	evidence_type	metric_name	observed_value	threshold	severity	plain_english_summary
							Severity: high.
ev_a07f9ac3f3b75c1484	obs_31f1dbac37c15eb7186d	annual_unit_value_change	unit_value_yoy_change	1.99837	0.8	high	2022 KWT to NPL 710812 (Non-monetary unwrought gold) has a year-over-year log unit-value change of 1.998. Severity: high.
ev_cb5cbcfcd460218ad	obs_31f1dbac37c15eb7186d	value_quantity_divergence	value_quantity_divergence	1.99837	0.8	high	2022 KWT to NPL 710812 (Non-monetary unwrought gold) shows value-quantity divergence of 1.998. Severity: high.
ev_7860c8fa03d5b08d13	obs_8c4051f3c18beab9e68a	history_deviation	robust_historical_z	19.2111	4	high	2022 QAT to NPL 710812 (Non-monetary unwrought gold) differs from its prior corridor history with robust historical z of 19.211. Severity: high.
ev_d6a6bd050796d6d0c2	obs_8c4051f3c18beab9e68a	benchmark_adjusted_drift	benchmark_adjusted_drift	1.77752	0.7	high	2022 QAT to NPL 710812 (Non-monetary unwrought gold) moved by 1.778 in benchmark-adjusted residual versus the previous observed corridor year. Severity: high.
ev_80601810fc818cf4e6		benchmark_gap	benchmark_residual	1.72365	0.7	high	2022 QAT to NPL 710812

evidence_id	obs_id	evidence_type	metric_name	observed_value	threshold	severity	plain_english_summary
	obs_8c4051f3c18beab9e68a						(Non-monetary unwrought gold) has a log benchmark residual of 1.724 against the World Bank annual benchmark. Severity: high.
ev_a600abc5ed980ad59a	obs_8c4051f3c18beab9e68a	annual_unit_value_change	unit_value_yearly_change	1.77806	0.8	high	2022 QAT to NPL 710812 (Non-monetary unwrought gold) has a year-over-year log unit-value change of 1.778. Severity: high.

Brief validation summary

obs_id	evidence_id_coverage	unsupported_numeric_claim_count	hallucinated_source_count	missing_caveat_flag	prohibited_language_flag	conclusion_boundary_present	typology_card_ids_present	status
obs_c1178b54e91326a9b0f5	True	0	0	False	False	True	True	pass
obs_31f1dbac37c15eb7186d	True	0	0	False	False	True	True	pass
obs_8c4051f3c18beab9e68a	True	0	0	False	False	True	True	pass
obs_c4e5ade4d38d2d3dd148	True	0	0	False	False	True	True	pass
obs_f7e56d6a0431cdbb8c32	True	0	0	False	False	True	True	pass

The GenAI layer is implemented as a GenAI-ready, evidence-grounded brief design. The verified output uses deterministic offline rendering; no live LLM call was made.

Key limitations

Purpose: Explain the project without overclaiming.

- The public BACI panel is annual and aggregate. It is not invoice, shipment, customer, payment, vessel, customs-document, or beneficial-ownership data.
- Commodity benchmarks are macro context and not invoice-level fair value.
- Synthetic review-priority scenarios are for ML evaluation only; they are not real TBML labels.
- FATF-Egmont materials are typology and caveat context only. They are not model features, row-level evidence, labels, or proof.
- Model scores are review-priority scores, not calibrated probabilities or legal findings.

Conclusion

Purpose: State exactly what the package supports.

The lab produces a validated official-data review queue, evidence table, deterministic analyst briefs, and static dashboard concept for human review prioritization. It supports consistent triage and documentation of unusual corridor-product-year patterns. It does not establish money laundering, misinvoicing, or criminal intent.